# VF Asia Footwear Laboratory Testing Manual

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| Date Issued : May 17, 2016<br>Revision # : | Created by : Nila Babia Erazo |

**PURPOSE AND SCOPE**: To determine the resistance of an assembly, sheet materials and components such as straps, and top lines to repeated stretching and flexing.

**REFERENCE:** SATRA TM 147

**EQUIPMENT and MATERIALS:**

1. Upper Material Stretch and Flex Machine – STM 117, meeting the requirements of SATRA TM 147. Refer to figure 4, flexing geometry.
2. Optical magnifier or stereoscopic microscope

3. Test Specimens, either of the 4 types :
   - Top line
   - Seam
   - Flat
   - Folded

**SAMPLE PREPARATION:**

***Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.***

1. Prepare the test sample, either cut from completed item of footwear or produced specifically for testing. Refer to figure 1 on different test specimens.

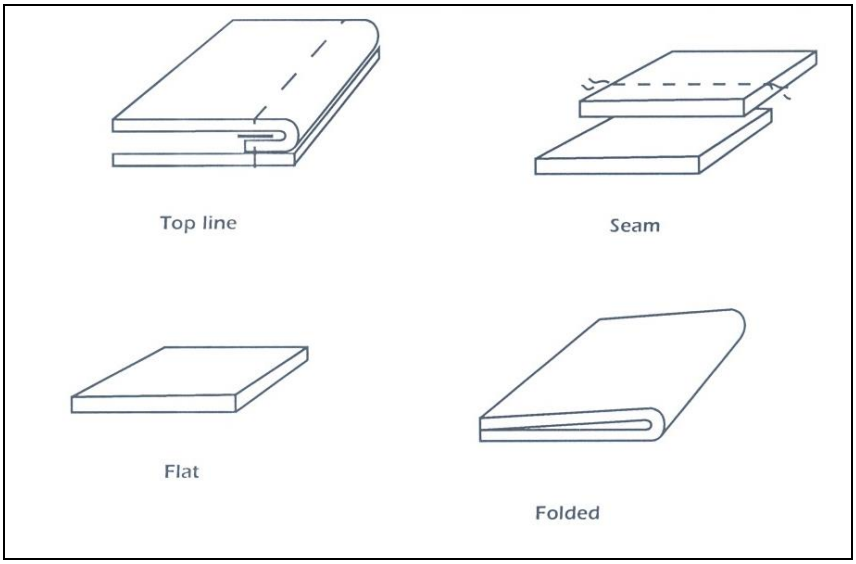



Figure 1

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- The completed test specimen should usually be a rectangle of width 25 ±2 mm and length 38 ± 2 mm. Thus , for example to produce a folded test specimen it will be necessary to cut a rectangle of width 50 ± 4 mm and length 38 ± 2 mm which can then be folded in half.

- When testing over seams, each end of the threads will need to be tied off, knotted and stuck together, so there is no possibility of them to unravelling during the test. For fabrics which are likely to fray or ladder from the edge, the test specimen should be slightly wider than usual up to a maximum of 35 mm and just the center portion clamped.

- For topline the test specimen should be 25±2 mm wide across the center of the specimen. The curvature of the topline will make the test specimen wider at the ends. The additional width should be left intact and protruding from the clamps. Topline test specimens are usually loaded with topline tensioned, see Figure 2

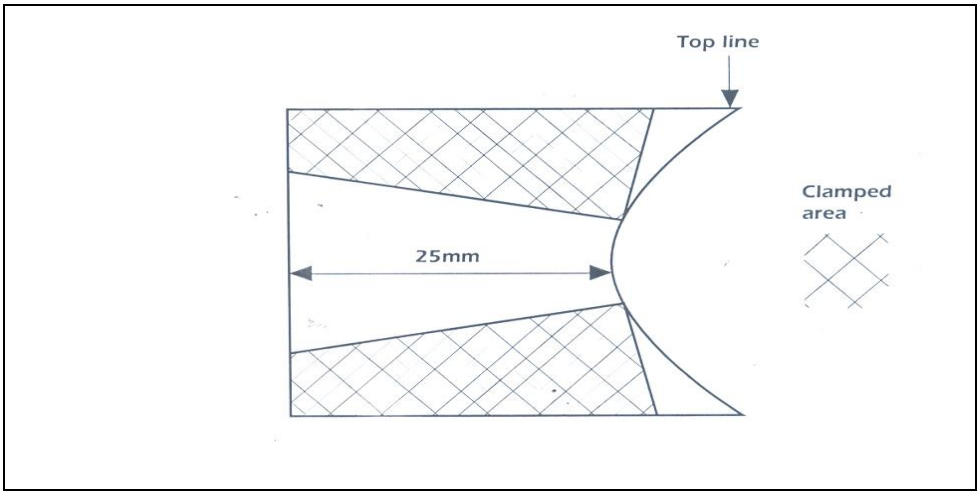



Figure 2

2. Draw lines on each test specimen to distinguish between the areas which will be held in the flexing machine clamps and the area which will be stretched and flexed, see Figure 3. Two methods of mounting the test specimen into the flexing machine can be used:

***I. STRAIGHT :***

- Draw a line across the width of the test specimen, through its center and parallel to its shorter edges.
- Draw a further two lines parallel to the center line, one 6.2 ±0.2 mm from either side of the center line.
- Mark one of these lines AC and the other BD.

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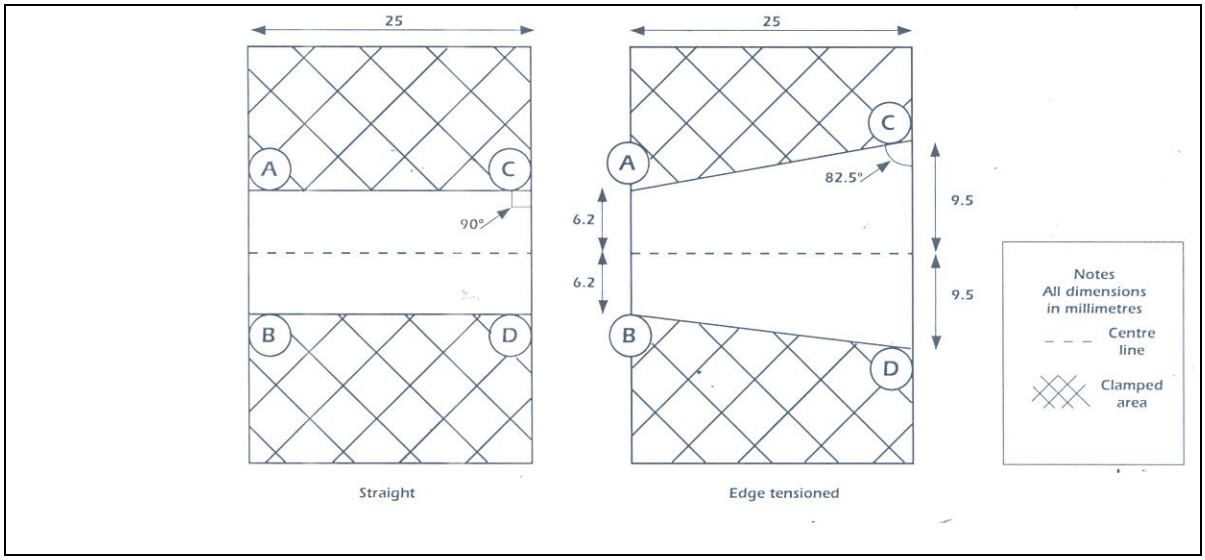



Figure 3: Marking test specimen

***II. EDGE TENSIONED :***

- Draw a line across the width of the test specimen, through its center and parallel to its shorter edges
- On one of the longer edges of the test specimens mark two points, one 6.2 ±0.2 mm from either side of the center line. Label one of the points A and the other B.
- On one of the longer edges of the test specimens mark two points, one 9.5 ±0.2 mm from either side of the center line. The point which is nearest to point A should be labelled as C and the other should be labelled as D.
- Draw a straight lines between the points A and C and the points B and D

**REPETITIONS:**

- From footwear uppers: test three (3) samples
- Sheet material: test three (3) samples for each direction

**TEST PROCEDURE:**

***Test should be conducted at standard laboratory environment @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.***

Load each test specimen into the flexing machine as follows:

1. Loosen the stationary clamp and the moveable clamp.

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2. Adjust the machine so that the clamps are as far as possible.

3. Place the end of the test specimen closest to the line AC into a moveable clamp and insert until the line AC is coincident with the front edge of the clamp.

4. Fully tighten the moveable clamp, while taking care not to overtighten it and damage the test specimen.

5. Adjust the machine so that the clamps move together while at the same time feeding the free end of the test specimen into the stationary clamp. If loading an edge tensioned test specimen it will also be necessary to buckle the longer edge of the specimen having the points C and D on it until the two lines AC and BD are approximately parallel.

6. Stop when the second line, marked BD, on the test is coincident with the edge of the stationary clamp. This should occur when the jaws are approximately 12.5 mm apart.

7. Fully tighten the stationary clamp, while taking care not to overtighten it and damage the test specimen.

8. Refer to figure 4 on flexing geometry.

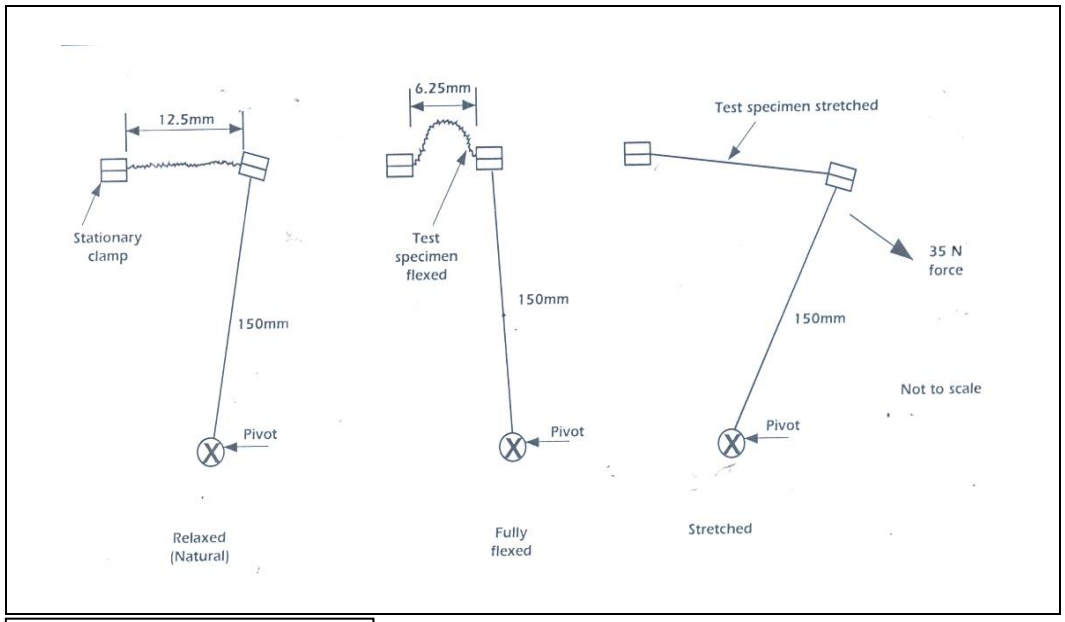



Figure 4: Flexing Geometry

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9. Reset all the counters to zero, and set the desired number of cycles.

10. Initiate testing, and ensure that the counter is operating for each pair of jaws loaded with test specimen.

11. Inspect the samples after every 5000 cycles during the first 20,000 cycles, after 20,000 cycles inspect the samples every 10,000 cycles.
    - At each inspection stage, assess the test specimen by using either the optical magnifier or the unaided eye.
    - When looking for damage inspect the specimen on the top surface and on the reverse side both while it is flexed and stretched.
    - Note any crease, cracking, splitting or tearing which has occurred.
    - Rate as slight, marked, or severe.

12. If the damage of the specimen is considered to be unacceptable, then remove the test specimen from the flexing machine and record the number of flexing cycles to which the specimen failed.

13. If failure has not occurred after the specified # of cycles, report the result as no failure.

**REPORTS:**

1. Appearance of any crease, cracking, splitting or tearing which has occurred.
2. Rate as slight, marked, or severe
3. Number of flexing cycles.
    - Calculate the arithmetic mean of the total number of flexing cycles for each type of test specimen.
    - For sheet materials calculate separate means for the along and across directions

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